Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/02/2019
PI Name:	Billman, Dorrit Ph.D.		
Project Title:	Training for Generalizable Skills & Knowledge: Integrating Principles and Procedures		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (3) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/06/2015	**End Date:**	07/05/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	3
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/5/2019 per NSSC information (Ed., 8/9/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Space Human Factors & Habitability (Ed., 1/19/17)		
Key Personnel Changes/Previous PI:	May 2017: no changes		
COI Name (Institution):	Catrambone, Richard Ph.D. (Georgia Tech Research Corporation)		
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Performance Goal No.:			
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	Understanding what properties of skills and knowledge (S&K) support useful generalization and long-term retention is critical to the success of future long-duration missions (e.g., DRM:Mars), yet there is a gap in our understanding of these properties (Space Human Factors Engineering-SHFE-TRAIN-04; ED. NOTE Sept 2019--now TRAIN-04: We do not know the types of skills and knowledge that can be retained and generalized across tasks for a given mission to maximize crew performance.). Broadly, it is often the case that specialized procedural skills may be well-retained but difficult to generalize and abstract declarative knowledge may support generalization but is difficult to retain. Nevertheless, findings from the literature suggest that integrated knowledge is likely to be both generalizable and well retained. By integrated skills and knowledge we refer to ensembles of a) procedural skills and b) knowledge of principles that are integrated through relations such as explanation, instantiation, and associated use. We will conduct literature research followed by piloting and experimentation to a) define appropriate measures and manipulations and b) assess the extent that integrated S&K generalizes well and is well-retained relative to S&K covering related content but without the integrating relationships. Experimentation will aim to produce integrated knowledge though a variety of learning activities that a) build links from procedural skills to abstract principles through processes such as self-explanation, thus facilitating their decomposition and reuse and b) build links from abstract principles to specific procedures through processes such as instantiation, thus aiding their application. Evidence shows that integrated knowledge is better retained, as the many links of integrated knowledge provide multiple, alternative retrieval cues accessible in many contexts. We will assess whether manipulations that produce integrated knowledge also produce better generalization (as well as measuring retention), relative to knowledge that spans related content but is not integrated, and whether such effects can be attributed to the mediating role of integrated knowledge.
Task Description:	We will use the work domain of operating and trouble-shooting complex technology, specifically spacecraft life-support systems as the test case for the proposed research. We will investigate “performing system-related tasks in highly autonomous environments” (Human Exploration Research Opportunities (HERO) Appendix A-25), such as operation, maintenance, and troubleshooting components within habitat life-support systems. Three factors motivate this choice.1) Such work is highly relevant to long-distance, crewed missions. 2) Prior research has studied operation of devices and thus provides findings on representations supporting generalization in similar contexts. 3) We will use an existing software suite including simulation of life-support devices on the International Space Station (ISS) and procedures for their operation as our test bed. This “micro-world” enables experimental control yet is a close analog of real mission work. In this domain the critical skills and knowledge are primarily cognitive (e.g., deciding what procedure to use), rather than sensory-motor (e.g., how hard a wrench must be turned). We include “meta-knowledge” such as identifying what information is missing when resources such as just-in-time-training are relevant. We will assess whether interventions related to integration produce forms of S&K that generalize well and are well retained, a prediction motivated by prior findings but not investigated directly, nor in NASA-relevant context. If integration is the basis for the predicted outcomes, it will provide a powerful principle for identifying and creating forms of S&K that are both generalizable and retainable. If not, we have discovered how learning is affected by important training interventions in a domain highly relevant to NASA future crewed exploration missions. This research will narrow the gap in understanding the factors that make skills and knowledge for NASA-critical tasks more generalizable and more retainable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Technical work across science, technology, engineering, and mathematics (STEM) disciplines, on Earth as in space, requires mastery of complex suites of knowledge and skills. Whether presented as education or as job-specific training, these skills and knowledge must be learned. Much of this work has an open, changing character such that even if time were available, it is not possible to anticipate and teach all the components that will be needed in the work. Thus, training that allows a person effectively to transfer skills and knowledge to situations and tasks that were not trained is extremely valuable. Indeed, transfer to different settings (from training environment to actual work) is often considered the primary success criterion for training; just a transfer from specific problem and procedures used in the classroom to novel situations is a critical goal of education. Understanding what types of knowledge and skills support transfer (and retention) will facilitate more effective training, for technical domains related to those we study.
Task Progress:	Our research continues investigation of training for effective generalization (transfer to new content). Our domain is equipment operation, specifically, operation of (simulated) habitat equipment on the International Space Station. We explore whether training that includes a model of the device being operated as well as experience running procedures for operation aids generalization to novel tasks, such as running procedures with an unexpected problem, writing a new procedure for a new goal, or distinguishing and explaining normal from nonnormal states. In the last year and a half since the last task book report we ran two experiments. We have also been working on a taxonomy of the factors, in training and in use, that affect successful generalization. We conducted an opportunistic study on retention of skills learned in the first two studies done under this grant. We contacted and arranged the return of previous participants--5 returning from Experiment 1 after a bit less than two years and 7 from experiment 2 returning after a bit less than one year. We recruited 14 new matched participants. We provided all participants with very basic (re) training on how to use the software, but minimal explanation or practice. Individual variation was large and overall performance was broadly similar. One of the key transfer tasks showed a significantly higher completion rate by returning users (6 of 12 vs 0 of 14) and the number of action-fails triggered by users was lower for returning than new users. The fourth study under this research grant compared a condition with integrated training of model and procedural knowledge to a condition only training procedural knowledge. We found that after training participants in the integrated model condition produced better diagrammatic and text explanations of how the Carbon Dioxide Removal Systems (CDRS) work than it did participants in the no-model condition. Thus, we did succeed in teaching the participants at least model information (whether or not integrated with procedural skill). A Monitoring Task asked participants to task discriminate and explain whether a statically presented situation is normal or problematic; participants in the Integrating Model Condition provided better explanations than did those in the No Model Condition. However, participants were similarly likely to produce appropriate, successful performance on procedure execution tasks that required recognition of a problem or potential problem. Across these studies we have also informally observed variety in behavior that seems to be influenced by a variety of other factors, both in training and in the transfer tasks. Identification and organization of these factors may be helpful in

	identifying factors limiting performance in a specific technical situation. We have investigated some of these factors but further understanding is needed.
Bibliography Type:	Description: (Last Updated: 10/03/2019)
Articles in Peer-reviewed Journals	Billman D, Catrambone R, Feldman J, Caddick Z, Eurich S, Leventhal J, Martin R, Sliwinska K. "Training for generalization: The role of integrated skills and knowledge in technology domains." Proceedings of the Human Factors and Ergonomics Society Annual Meeting. 2018 Sep;62(1):1434-8. (62nd Annual Meeting of the Human Factors and Ergonomics Society, Philadelphia, Pennsylvania, October 1–5, 2018.) https://doi.org/10.1177/1541931218621326 , Sep-2018